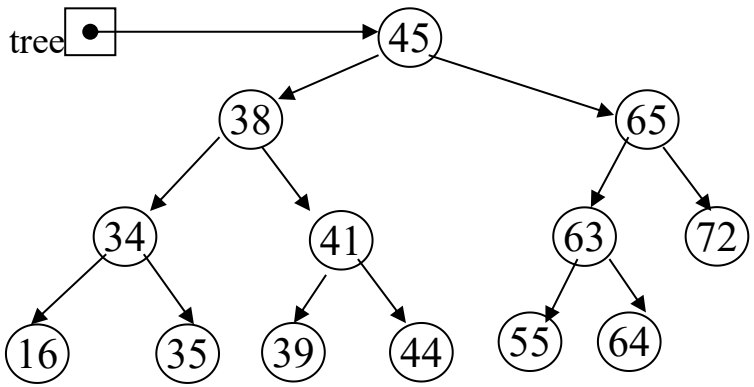
1. Given the following binary tree:



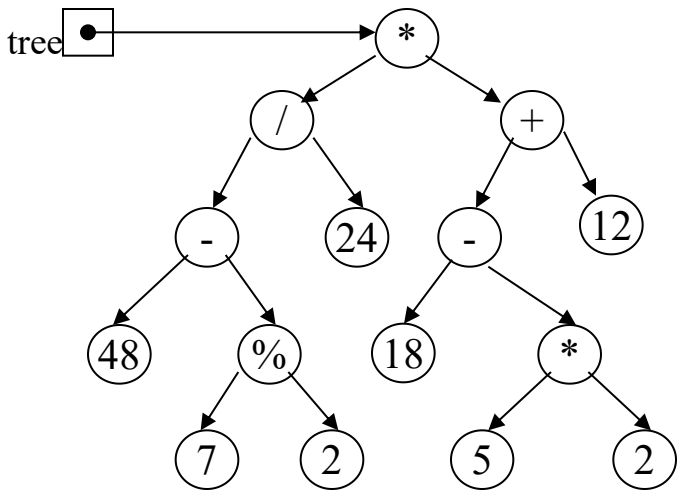
(a) What is the inorder traversal of the tree? (left -> root -> right)
19, 34, 35, 38, 39, 41, 44, 45, 55, 63, 64, 65, 72

(b) What is the preorder traversal of the tree? (root-> left-> right)
45, 38, 34, 16, 35, 41, 39, 44, 65, 63, 55, 64, 72

(c) What is the postorder traversal of the tree? (left -> right-> root)
16, 35, 34, 39, 44, 41, 38, 55, 64, 63, 72, 65, 45

(d) What is the height of the tree? What nodes are on level 2? (Start at 0)
Height = 3. Nodes = 34,41, 63, 72

2. Given the following binary expression tree:



(a) What is the inorder traversal of the tree?

48 - 7 % 2 / 24 * 18 - 5 * 2 + 12

(b) What is the postorder traversal of the tree?

48, 7, 2, %,-, 24, /, 18, 5, 2, *, -, 12, +, *

(c) What does it evaluate to if using integer division?
20

(d) What does it evaluate to if using float division?

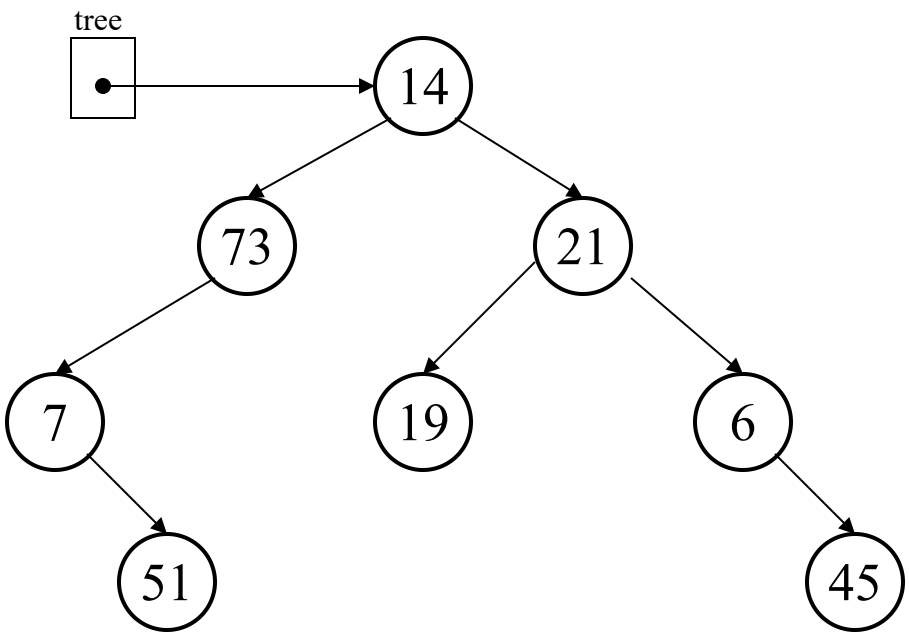
39.17

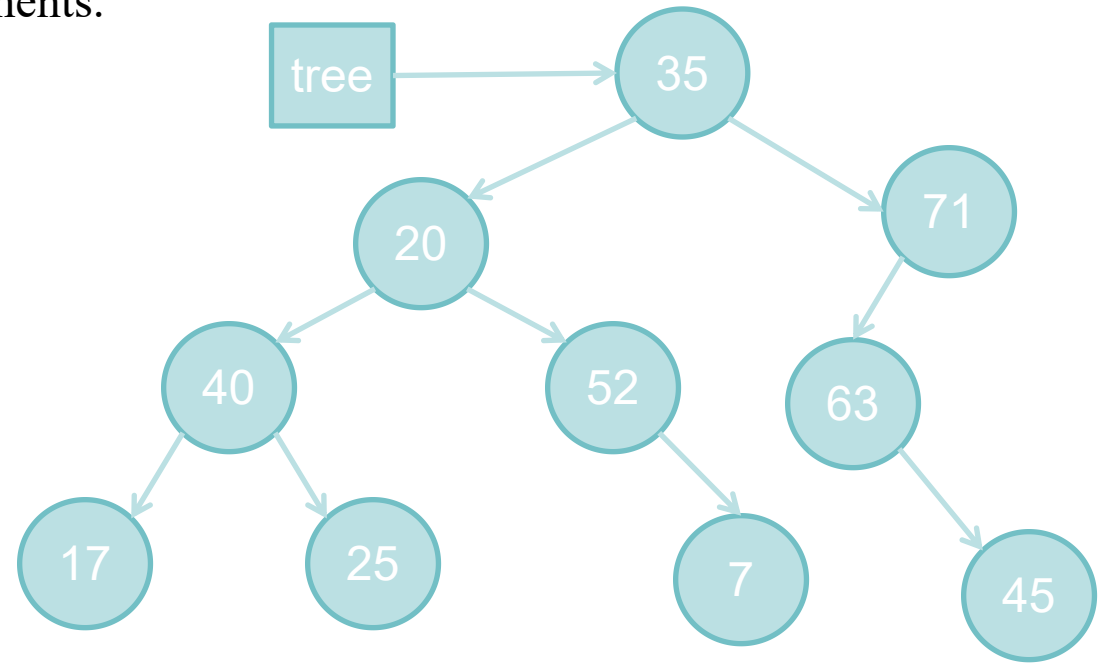
3. The elements in a binary tree area to be stored in an array. Each element is a nonnegative int value.

a. What value can you use as a dummy value, if the binary tree is not complete? -1 or IN

b. Show the contents of the array, given the tree illustrated below

[0]	14
[1]	73
[2]	21
[3]	7
[4]	-1
[5]	19
[6]	6
[7]	-1
[8]	51
[9]	-1
[10]	-1
[11]	-1
[12]	-1
[13]	-1
[14]	45



4. Given the array pictured below, draw the binary tree that can be created from its elements.

[0]	35
[1]	20
[2]	71
[3]	40
[4]	52
[5]	63
[6]	null
[7]	17
[8]	25
[9]	null
[10]	7
[11]	null
[12]	45